

Q4(a)(i)

When the waves arrive at the point of observation with a path difference of $n\lambda$, where $n = 0, 1, 2, 3, \dots$, they interfere constructively to give a loud note.

When the waves arrive at the point of observation with a path difference of $(n + \frac{1}{2})\lambda$, they interfere destructively to give a soft note.

Q4(a)(ii)

Using Young's double slit formula,

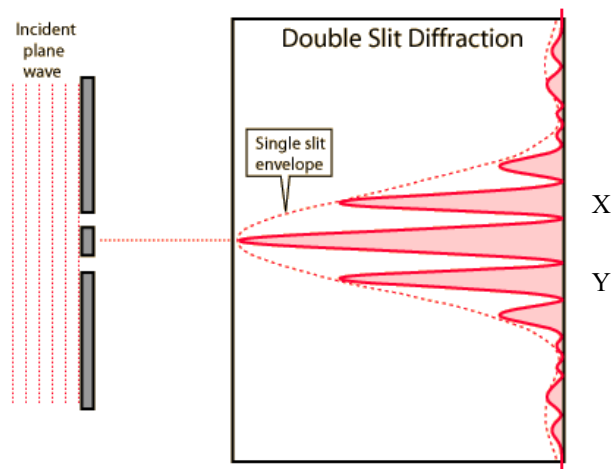
the distance between two adjacent loud regions is $y = \frac{\lambda L}{d}$.

$$y = \frac{(4.8 \times 10^{-2})(10)}{75 \times 10^{-2}} = 0.64 \text{ m}$$

Hence the distance between an adjacent loud and soft regions is $0.64/2 = \mathbf{0.32 \text{ m}}$

Q4(a)(iii)

The loudspeakers behave like double slits.



Waves from the double slits interfere to produce loud and soft regions at the points of observation.

Single slit diffraction from the aperture of a loudspeaker (each) forces the intensity regions to be confined within a diffraction envelope.

This produces an intensity profile that increases from X to centre and thereafter decreases to Y.

Q4(b)(i)

If the diameter d of the loudspeaker is smaller than the wavelength λ of sound, the latter will be able to diffract effectively through the loudspeaker.

Q4(b)(ii)

The wavelength of sound is $\lambda = \frac{v}{f}$.

The wavelengths of notes from the piano range from 340/3400 to 340/34 m or 0.10 m to 10 m.

Hence $d < \lambda = \mathbf{0.10\ m}$ will ensure adequate spreading for all wavelengths.

Q5(a)